

EXP 1

Objective 1

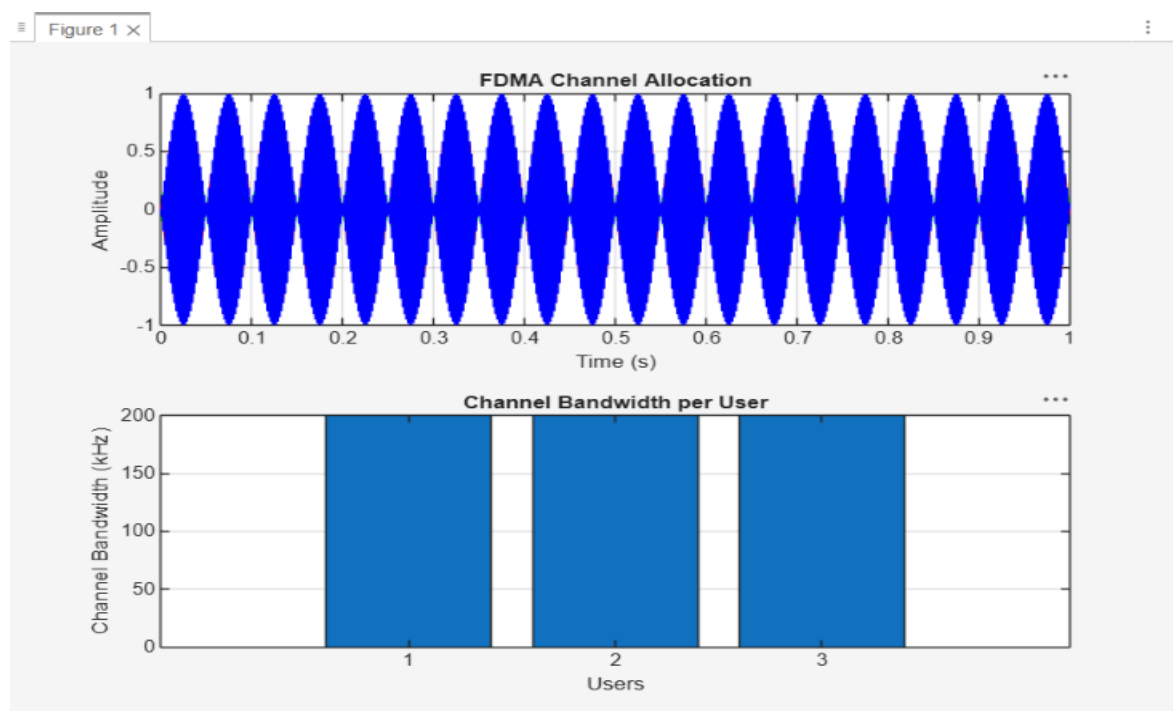
Program

```
clc; clear; close all;
fs=1000; t=0:1/fs:1;
TBW=600; Users=3;

CBW=TBW/Users;
fc=[100 300 500];
m=sin(2*pi*10*t);
s1=m.*cos(2*pi*fc(1)*t);
s2=m.*cos(2*pi*fc(2)*t);
s3=m.*cos(2*pi*fc(3)*t);
subplot(2,1,1)
plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on
title('FDMA Channel Allocation'),xlabel('Time (s)'),ylabel('Amplitude')
subplot(2,1,2)
bar(1:Users,CBW*ones(1,Users)),grid on

xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per User')
```

OUTPUT

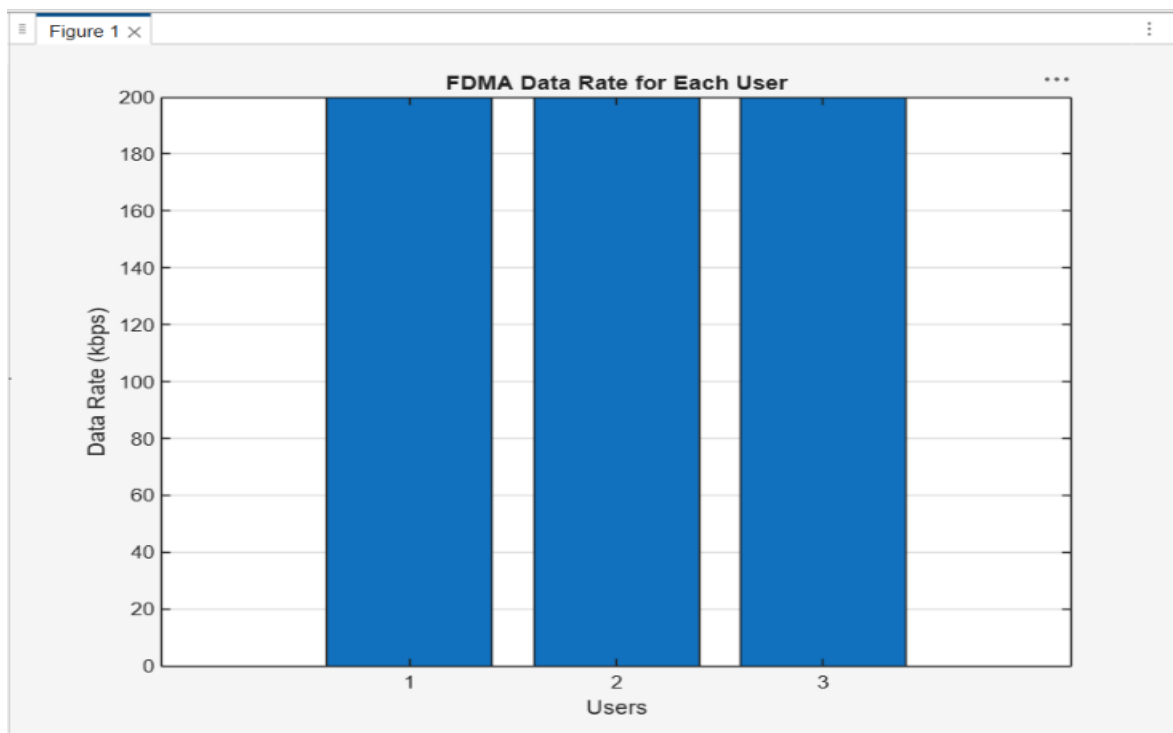


Objective 2

Program

```
clc; clear; close all;
BW = 200;
Users = 3;
M = 2;
DataRate = BW*log2(M);
Rate = DataRate*ones(1,Users);
disp(' User  Data Rate
(kbps)') disp([(1:Users)'
Rate'])
figure
bar(1:Users,Rate)
e)
xlabel('Users')
ylabel('Data Rate (kbps)')
title('FDMA Data Rate for Each
User') grid on
```

OUTPUT



Objective 3

```
clc; clear; close all;
Ps = 1;
Pn = [0.1 0.05 0.01];
Users = 1:3;
SNR = 10*log10(Ps./Pn);
disp('User Noise Power(W) SNR(dB)')
disp([Users' Pn' SNR'])
figure
bar(Users,SNR)
xlabel('Users')
ylabel('SNR (dB)')
title('Signal-to-Noise Ratio of FDMA Users')
grid on
```

Output

